

# Matter, Solitons, and Emergent Mass in a Deterministic Brane-Lattice Substrate (Paper IV)

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## Abstract

We develop the matter and mass consistency bridges for the deterministic 4D brane-lattice substrate of Paper I. The matter bridge poses particle-like states as self-adjoint eigenproblems with angular structure organized by vector spherical harmonics (VSH). Two mechanisms together establish Derrick stability: anti-collapse is provided by the lattice spacing  $a$  (no continuous Derrick rescaling below one cell), and anti-expansion by the geometric quartic  $\propto k_s \alpha / a$ , giving an equilibrium soliton radius  $R_H / a \approx \kappa(A/a) \sqrt{\alpha / (1 - \alpha)}$  with a soliton sweet spot at  $\alpha \approx 0.5\text{--}0.8$ . The canonical baryon ansatz is the hedgehog  $\xi^i = f(r) \hat{x}^i$  ( $J = 0, L = 1$  VSH), which lives entirely in the  $SU(3)$  traceless sector and carries neutron-like far-field holonomy; a proton-like far field requires a nonzero  $L = 0$  scalar admixture. Soliton states carry labels  $(J, P; SU(3)\text{-irrep}; Q_{U(1)}; B_{\text{winding}})$  with charge-triality locking  $q \equiv -t \pmod{3}$  inherited from Paper III. Periodic boundary conditions are mandatory: open BC allows the prestressed lattice to relax (contracting to  $\alpha a$ ), collapsing even the  $A = 0$  vacuum. The mass bridge derives rest mass from self-confinement, not from the Higgs mechanism: a wave permanently trapped in a closed-loop toroidal mode has a total substrate energy  $E(R^*, A) = mc^2$  at the equilibrium major radius  $R^* = \sqrt{E_{\text{curv}} / (2\pi T_{\text{string}})}$ . The complex structure  $\Psi \approx \xi + (i/\omega_0)\dot{\xi}$  originates entirely in the timelike worldtube direction; the  $U(1)$  carrier frequency  $\omega_0$  is fixed by the worldtube closure condition. Time-link prestress sources a binding compression  $\Delta u_{\parallel}^{\text{ext}} \propto -\alpha \omega^2 < 0$  that flips the  $U(1)\text{--}SU(3)$  coupling to attractive when  $\chi = 2\alpha(\omega/\omega_*)^2 (A_0/a)^2 > 1$ . Both bridges are gated on the same object: a converged stable baryon-like eigenstate, the decisive numerical target of the series.

**Keywords:** soliton; Derrick stability; vector spherical harmonics; hedgehog ansatz; Skyrme texture; emergent mass; self-confinement; toroidal mode; time-link binding; brane lattice

## 1. Introduction

This is the fourth and final paper in the series. Paper I established the substrate model. Papers II and III developed the Lorentz, gravity, gauge, and color bridges. The present paper develops two bridges that have been deliberately held back: matter and mass.

**Matter bridge.** The matter bridge poses particle-like states as self-adjoint eigenproblems on a symmetric background, with angular structure organized by vector spherical harmonics (Section 3). The key structural inputs are the geometric quartic  $W_4 \propto k_s \alpha / a$  from Paper I and the  $U(3)$  triplet structure from Paper III. Two mechanisms together establish stability: (i) anti-collapse is provided by the lattice spacing  $a$  — the Derrick  $\lambda \rightarrow 0$  rescaling halts at one cell, because configurations concentrated below one cell are not local energy minima of the spring network; (ii) anti-expansion is provided by the geometric quartic  $\propto k_s \alpha / a$ , which

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under Derrick rescaling scales as  $\lambda^{-1}$  and overpowers the  $\lambda^{+1}$  gradient term for  $\lambda > \lambda^*$ . The equilibrium soliton radius from this balance is  $R_h/a \approx \kappa (A/a) \sqrt{\alpha/(1-\alpha)}$ , setting a soliton sweet spot at  $\alpha \approx 0.5\text{--}0.8$ .

**Mass bridge.** The mass bridge derives emergent rest mass from self-confinement of a closed-loop wave mode — not from the Higgs mechanism (Section 4). A wave trapped in a circularly closed toroidal excitation cannot disperse: it is pinned by topology and geometric self-guidance. The total substrate energy of this mode at equilibrium,  $E(R^*, A)$ , is identified with the rest mass  $mc^2$ . The complex structure  $\Psi \approx \zeta + (i/\omega_0)\dot{\zeta}$  that mediates this picture arises entirely from the timelike worldtube direction — the imaginary unit  $i$  here is a quarter-period time shift, not an independent complex field. A separate, quantitative open problem is connecting the equilibrium radius  $R^*$  to the Compton wavelength  $\hbar/mc$ ; this requires deriving  $\hbar$  and  $c$  from substrate parameters consistently.

**Why matter and mass are paired.** Both bridges are gated on the same object: a converged, stable, localized soliton eigenstate. Without such an eigenstate, the mass identification  $E(R^*, A) = mc^2$  remains schematic. The matter bridge's decisive numerical task (does the substrate support a stable baryon-like  $\pi_3$  texture?) is therefore simultaneously the gate that unlocks the mass bridge's normalization argument. Pairing them in one paper makes this dependency explicit.

**Paper organization.** Section 2 lists imports from Papers I and III. Section 3 develops the matter bridge: self-adjoint eigenproblem, VSH angular structure, Derrick stability, soliton labels, and periodic-BC requirement. Section 4 develops the mass bridge: self-confinement picture, toroidal mode energy, Derrick balance for torus geometry, time-link binding, and open normalization problems.

## 2. Interfaces

This paper imports the following results from Papers I and III without re-derivation, and exports the localized-mode results consumed by Paper V.

*From Paper I*

Equations of motion. Equation (26).

Geometric nonlinearity and quartic coefficient. The constitutive quartic  $W_4 \propto k_s \alpha / a$  (Equation (20)), which provides the anti-expansion stabilization of any localized mode.

Induced metric and near-identity expansion. Equations (10) and (29); the transverse–lateral coupling  $\partial_i u \partial_j u$  underlies both the gravity channel (Paper II) and the geometric self-guidance discussed here.

Continuum action. Equation (24).

Discrete lattice action. Equation (7); the lattice spacing  $a$  is the UV cutoff that blocks Derrick collapse.

Bell interpretive stance. Section 5 of Paper I.

*From Paper III*

$U(3)$  triplet decomposition. The axis-aligned lateral triplet  $\Psi \in \mathbb{C}^3$ , its  $\mathfrak{u}(3) = \mathfrak{u}(1) \oplus \mathfrak{su}(3)$  decomposition, and the  $U(1)$  trace identification with electromagnetism (Paper III Equations (13) and (14)).

Kinematic color confinement. The result that color charge cannot be carried to infinity by a free linear excitation (Paper III Section 4.4).

Charge-triality locking.  $q \equiv -t \pmod{3}$ , which constrains the soliton labels defined in Section 3.4.

Gauge-covariant envelope dynamics. The covariant derivative  $\mathcal{D}_\mu \Psi = (\partial_\mu + i\mathcal{A}_\mu)\Psi$  and effective action  $S_{\text{eff}}$  (Paper III Equations (10) and (11)).

*Exported to Paper V*

Localized soliton modes and chirality. The localized matter modes (Section 3) and their forward/backward propagation along the timelike axis, which Paper V (Bell/interpretation) reads as matter and antimatter worldtubes.

### 3. Matter bridge: solitons as self-adjoint eigenproblems

#### 3.1. Linearization about symmetric backgrounds

Let  $\mathbf{X}_0(x)$  be a time-independent background embedding representing a localized deformation region, taken to be spherically symmetric in the intrinsic brane coordinates ( $r = |x|$  with  $r \gg a$  for the continuum limit to apply). Linearize the equations of motion (Equation (26) of Paper I) about  $\mathbf{X}_0$  by writing  $\mathbf{X}(x, t) = \mathbf{X}_0(x) + \boldsymbol{\eta}(x, t)$  with small perturbation  $\boldsymbol{\eta}$ . Because the dynamics derives from the action (Equation (24) of Paper I), the linearized operator is the second variation of  $S$  and is therefore self-adjoint with respect to the  $\rho_m$ -weighted  $L^2$  inner product on  $\Omega$ ; orthogonality and completeness of eigenmodes follow as standard consequences.

#### 3.2. Angular structure: vector spherical harmonics

The substrate is the 6-neighbor cubic lattice, whose point group is  $O_h$  rather than  $SO(3)$ . At scales  $\gg a$ , however, long-wavelength continuum modes recover approximate isotropy with residual cubic corrections of order 7.5% in longitudinal speed at  $\alpha = 0.2$ . Solitons in this theory are expected to live in the geometric-quartic-stabilized regime with  $\ell_{\text{soliton}} \gg a$ , so the emergent approximate  $SO(3)$  is the right symmetry to organize their angular structure, with  $O_h$  corrections entering as small splittings of  $SO(3)$  multiplets.

For a spherically symmetric background the linearized operator commutes with the action of  $SO(3)$  on the intrinsic coordinates at leading order in  $a/\ell_{\text{soliton}}$ . For a scalar field this gives the familiar separation into radial and angular problems with eigenfunctions  $Y_{\ell m}(\theta, \varphi)$ . The lateral triplet  $\zeta^i \in \mathbb{R}^3$  and its complex-envelope promotion  $\Psi \in \mathbb{C}^3$  from Paper III carry an internal vector index  $i$  in addition to spatial dependence. The correct angular basis is therefore the *vector spherical harmonic* (VSH) basis  $\mathbf{Y}_{JLM}^i$ , where  $J$  is total angular momentum,  $L$  is orbital angular momentum, and  $L \in \{J-1, J, J+1\}$ :

$$\zeta^i(r, \theta, \varphi) = \sum_{J,L,M} f_{JLM}(r) \mathbf{Y}_{JLM}^i(\theta, \varphi). \quad (1)$$

The choice of  $(J, L)$  specifies how the internal index  $i$  is glued to the spatial angular coordinates — it is the choice of how the triplet's  $U(3)$  gauge structure (Paper III) couples to the geometry of the localized mode. Configurations with  $J = 0$  are spherically symmetric only under the *combined* rotation that acts on both spatial and internal indices together; this combined rotation is the physically meaningful notion of spherical symmetry for a triplet field.

#### 3.3. Hedgehog ansatz and Skyrme-twisted extension

The canonical baryon ansatz is the *hedgehog*

$$\zeta^i(\mathbf{x}) = f(r) \hat{x}^i, \quad (2)$$

with  $J = 0, L = 1$ : the internal index  $i$  is locked to the spatial direction  $\hat{x}^i$ , so the configuration is invariant only under the combined diagonal  $SO(3)$ . The entire nontrivial

structure of the hedgehog lives in the  $SU(3)$  traceless sector of Paper III: the  $U(1)$  trace  $f(r) (\hat{x}^1 + \hat{x}^2 + \hat{x}^3)$  averages to zero on the sphere, giving a trace-neutral far field (neutron-like far-field holonomy).

Topological stabilization at winding number  $B = 1$  is achieved by extending the hedgehog into a *Skyrme-twisted* configuration in which the embedding-amplitude direction  $X^4$  also rotates radially:

$$(\xi^i, u)(\mathbf{x}) = (f(r) \sin F(r) \hat{x}^i, f(r) \cos F(r)), \quad (3)$$

with the profile  $F : [0, \infty) \rightarrow [\pi, 0]$  enforcing the boundary conditions  $F(0) = \pi$  and  $F(\infty) = 0$ . This boundary condition gives  $B = 1$  topological winding in  $\pi_3(SU(3)) = \mathbb{Z}$ , consistent with Paper III Section 4.3. Adding a nonzero  $L = 0$  scalar admixture shifts the  $U(1)$  far-field holonomy from trace-neutral to proton-like.

### 3.4. Soliton labels

A localized 3D mode is fully classified at the soliton layer by the labels

$$(J, P; SU(3)\text{-irrep}; Q_{U(1)}; B_{\text{winding}}), \quad (4)$$

where:

- $J$  is total angular momentum from the emergent approximate  $SO(3)$  at scales  $\gg a$ ; it refines to an  $O_h$  irrep at sub-leading order in  $a/\ell_{\text{soliton}}$ .
- $P$  is parity, the discrete element of the emergent rotation group.
- The  $SU(3)$  irrep and  $U(1)$  charge come from the  $U(3)$  decomposition (Paper III Section 4.2).
- $B_{\text{winding}}$  is the topological winding number in  $\pi_3(SU(3)) = \mathbb{Z}$ .

These labels are not independent: the VSH coupling sets a Clebsch–Gordan relationship between  $SO(3)$  and  $U(3)$  representations, and the charge-triality locking  $q \equiv -t \pmod{3}$  (Paper III Section 4.2) constrains  $Q_{U(1)}$  given the  $SU(3)$  irrep.

### 3.5. Stability: Derrick's theorem on the discrete substrate

In 3D, a standard Derrick scaling argument [1] rules out stable finitely-extended solitons in a purely quadratic (linear) theory: under  $x \mapsto \lambda x$ , the gradient energy  $E_2$  scales as  $\lambda^1$  (favoring expansion for  $\lambda > 1$ ) while a quartic energy  $E_4$  scales as  $\lambda^{-1}$  (resisting expansion). Without  $E_4 > 0$ , the soliton expands without bound. The present substrate breaks the standard Derrick counting in two complementary ways.

**Anti-collapse: lattice UV cutoff.** Derrick's argument assumes continuous rescaling  $\lambda \in (0, \infty)$ . On a discrete lattice with spacing  $a$ , however,  $\lambda$  is bounded below: a soliton cannot shrink past one cell. A configuration with energy fully concentrated at a single node is not a local energy minimum — the spring network immediately disperses it outward. The lattice spacing  $a$  is the physical UV cutoff; no separate quartic repulsion is needed to block collapse. This is a genuinely discrete-substrate effect that has no direct continuum analogue.

**Anti-expansion: geometric quartic.** Under Derrick rescaling in 3D, the geometric quartic  $W_4 \propto k_s \alpha / a$  (Paper I Equation (20)) scales as  $\lambda^{-1}$ . Balancing  $\lambda E_2$  against  $\lambda^{-1} E_4$  gives equilibrium at

$$\lambda^* = \sqrt{E_4/E_2}, \quad (5)$$

translating to an equilibrium soliton half-radius

$$\frac{R_h}{a} \approx \kappa \frac{A}{a} \sqrt{\frac{\alpha}{1-\alpha}}, \quad (6)$$

where  $\kappa$  is a numerical factor of order unity fixed by the Derrick balance in the specific ansatz and  $A$  is the peak amplitude. At  $\alpha = 0.2$  and  $A/a \approx 10$  this gives  $R_h/a \approx 5$ ; the soliton sweet spot  $\alpha \approx 0.5$ – $0.8$  gives  $R_h/a \approx 8$ – $14$  for the same amplitude, well above the lattice scale.

Note that the quartic coefficient scales as  $\propto \alpha/a$ , not  $\propto (1-\alpha)$  as a naive StVK identification would give. The correct sign and scaling are derived from the spring's geometric quartic in Paper I (Equation (20)), which inverts the naive StVK prediction. Quantitative soliton sizing must use the spring law, not StVK.

### 3.6. Periodic boundary conditions: mandatory requirement

Periodic boundary conditions (BC) are mandatory for all nonlinear soliton simulations. The prestressed lattice (rest length  $\alpha a < a$ ) is under tension; open/free BC allows the box to contract from  $Na$  to  $N\alpha a$ , which dominates all dynamics and collapses even the  $A = 0$  vacuum. Periodic BC fixes the box at  $N \cdot a$ , holds the tension, and ensures the vacuum is stable ( $A = 0 \Rightarrow E_{\text{excess}} \equiv 0$ ). All retracted 2026-06-05 runs used open BC; all future nonlinear runs must use periodic-clamped BC.

### 3.7. Matter bridge open problems

1. **Converged baryon eigenstate.** Obtain a converged, stable, localized  $\pi_3$ -winding solution using the periodic-clamped tensioned vacuum and an eigenstate approach (not seed-and-watch evolution). The hedgehog/Skyrme-twisted ansatz (3) with  $\alpha \approx 0.7$ ,  $A/a \approx 10$  is the primary candidate.
2. **Rigorous Derrick stability proof.** State Derrick's theorem precisely [1,2] for the present action, give the lattice-UV anti-collapse argument as a theorem (not just intuition), and tie the anti-expansion balance (5) to the effective field manifold of Paper III.
3. **Converged  $U(1)$  vortex eigenstate.** Resolve the semilocal drift observed in the preliminary JFNK run (Paper III Section 4.5) with branch-selection machinery for the semilocal vortex.
4. **Two-soliton interaction potential.** Once individual stable eigenstates exist, measure the two-soliton interaction potential as a function of separation and polarization alignment, to test whether the geometric self-guidance mechanism produces an attractive or repulsive channel.
5. **Spin- $\frac{1}{2}$  from soliton holonomy.** The spinorial  $4\pi$  aspect identified kinematically in Paper III (Section 3.6) must be realized by an actual localized mode: show that a converged soliton carries a  $\mathbb{Z}_2$  holonomy ( $\pi_1(SO(3)) = \mathbb{Z}_2$ ) under a  $2\pi$  spatial rotation, so that spin- $\frac{1}{2}$  emerges at the soliton level rather than from the linear envelope.

## 4. Mass bridge: rest mass from self-confinement

### 4.1. Physical picture: wave trapped in a closed loop

Rest mass in this framework is not acquired through the Higgs mechanism (spontaneous symmetry breaking and a scalar condensate). Instead, rest mass is identified with the total substrate energy of a wave mode that is permanently trapped in a circularly closed, self-confined loop:

$$mc^2 = E[\mathbf{X}_{\text{torus}}] \Big|_{R=R^*}, \quad (7)$$

where  $\mathbf{X}_{\text{torus}}$  is the toroidal mode embedding and  $R^*$  is the equilibrium major radius. The wave cannot disperse because it is topologically pinned (the  $U(1)$  vortex carries  $\pi_1(U(1)) = \mathbb{Z}$  winding) and geometrically self-confined (the geometric quartic prevents radial expansion). This is closer in spirit to the toroidal-electron proposal of Williamson and van der Mark [3], which attributes the correct gyromagnetic ratio  $g = 2$  to a circularly self-confined photon-like mode, than to the standard Higgs picture. The substrate grounds that proposal in a concrete elastic medium.

The proposal that rest mass is field or wave self-energy rather than an intrinsic attribute has a long pedigree. The electromagnetic-mass programme of Thomson, Heaviside, Abraham, Lorentz, and Poincaré [4–8] derived the inertia of the electron from the energy and momentum of its own field, and Einstein’s inertia–energy relation [9] established that the inertia of any body equals its rest energy divided by  $c^2$ . A modern wave-mechanical version is due to Chang [10]: particles as confined vacuum excitations with dispersion  $\omega^2 = (k^2 + l^2)c^2$  and rest mass set by a transverse confinement wavenumber,  $m_0 = \hbar l / c$ . The present construction realizes the same idea in a concrete elastic substrate, with  $l$  the inverse equilibrium loop radius  $l = 1/R^*$ , so that  $m_0 = \hbar l / c$  coincides with the Compton closure  $R^* = \hbar / mc$  of Section 4.5.

#### 4.2. Complex structure from the time direction

The complex structure  $\Psi \in \mathbb{C}^3$  used throughout this series is not an independent complex field. It arises from the carrier rotation of a real substrate field along the timelike worldtube axis. For a band-isolated carrier at frequency  $\omega_0$ , the complex envelope is defined by

$$\Psi \approx \xi + \frac{i}{\omega_0} \dot{\xi}, \quad (8)$$

where  $\xi$  is the real lateral triplet and  $\dot{\xi}$  is its time derivative. The imaginary unit  $i$  here is the quarter-period time shift (the time-evolution generator  $i\partial_t \Psi = H_{\text{eff}} \Psi$ ); it is not an independent degree of freedom. A single spacelike slice is a real snapshot and carries no phase; the  $U(1)$  phase exists only across the time extent of the worldtube. Two adjacent time slices are the minimum data that fix the  $U(1)$  — exactly the structure of the rotating-frame-periodic BVP solver that must be used for toroidal mode eigenstates.

This origin of the complex structure has a direct consequence for mass: the  $U(1)$  carrier frequency  $\omega_0$  is the rate of rotation of the real field along the time axis, and the energy stored in one carrier cycle is the substrate analogue of a quantum of energy. The mass–frequency relation  $E \approx \hbar \omega_0$  (in appropriate units) is therefore expected to hold at the equilibrium amplitude, with  $\hbar$  emerging from the substrate normalization (Section 4.5).

#### 4.3. Toroidal mode energy and Derrick balance

A toroidal mode is a  $U(1)$  vortex (line vortex in the trace sector) wrapped into a closed loop of major radius  $R$  and core radius  $w$ . The total substrate energy decomposes into gradient and quartic contributions:

$$E(R, A) = E_2(R, A) + E_4(R, A), \quad (9)$$

where at leading order in  $w/R$ :

$$E_2(R, A) \sim 2\pi R \cdot \varepsilon_2(A, w), \quad (10)$$

$$E_4(R, A) \sim 2\pi R \cdot \varepsilon_4(A, w), \quad (11)$$

with  $\varepsilon_2$  and  $\varepsilon_4$  the gradient and quartic energy densities per unit length of the vortex core. Both scale linearly with  $R$  (circumference grows), so to leading order the total energy is



linear in  $R$  — this is the string-tension picture. The correction that selects a finite equilibrium  $R^*$  comes from the curvature energy of the torus: turning the line vortex into a closed loop costs an extra curvature term  $\propto 1/R$ , balancing the string-tension term  $\propto R$ :

$$E(R) \approx T_{\text{string}} \cdot 2\pi R + \frac{E_{\text{curv}}}{R}, \quad (12)$$

with  $T_{\text{string}}$  the vortex string tension set by  $\varepsilon_2 + \varepsilon_4$  and  $E_{\text{curv}}$  the curvature energy from bending the vortex line. Setting  $dE/dR = 0$  gives the equilibrium radius

$$R^* = \sqrt{\frac{E_{\text{curv}}}{2\pi T_{\text{string}}}}. \quad (13)$$

The physical rest mass is then  $mc^2 = E(R^*) = 2 \cdot 2\pi R^* T_{\text{string}}$ : the gradient and curvature contributions are equal at equilibrium.

#### 4.4. Time-link prestress as binding agent

The timelike link in the discrete 4D brane action (Equation (7) of Paper I) carries a prestress  $r_t = \alpha\beta\Delta t$  analogous to the spatial prestress parameter  $\alpha$ . Expanding the time-link norm term  $-k_t r_t |\Delta R_t|$  produces a negative, lump-co-located energy density in the carrier velocity field:

$$\bar{E}_{t,\perp}(\rho) = -\frac{m\alpha}{4} (\omega A(\rho))^2 < 0, \quad (14)$$

where  $\omega$  is the carrier frequency fixed by the worldtube closure condition  $\omega T = 2\pi n$  and  $A(\rho)$  is the amplitude profile. This induces a definite-negative longitudinal compression on the spatial links:

$$\Delta u_{\parallel}^{\text{ext}}(\rho) = -\frac{m\alpha}{4k_s b^2} (\omega A(\rho))^2 \propto -\alpha\omega^2, \quad (15)$$

peaked at the vortex core  $\rho \approx w$ , zero on the axis. This is externally sourced by the time-link prestress and carrier eigenvalue, not slaved to the spatial relaxation.

Feeding (15) into the separable spatial cubic  $V_3 = +(k_s\alpha/2a^2)\Delta u_{\parallel}(\xi_s^2 + |\xi_{\perp}|^2)$  flips the net cross-coupling:

$$g_{\text{net}} = \frac{k_s\alpha}{4a^2} (1 - \chi), \quad \chi = 2\alpha \frac{\omega^2}{\omega_*^2} \frac{A_0^2}{a^2}, \quad \omega_* \equiv \frac{c_L}{b}. \quad (16)$$

*Binding* ( $g_{\text{net}} < 0$ , restoring force between  $U(1)$  and  $SU(3)$  sectors) holds iff  $\chi > 1$ . This is reachable at the soliton sweet spot  $\alpha \approx 0.5\text{--}0.8$ ,  $\omega$  near the band edge,  $A_0/a \approx 0.3$ . The spatial-only verdict (net repulsive) is the  $r_t \rightarrow 0$  limit; the time link genuinely flips the sign.

**Linchpin: closure-locked carrier frequency.** The binding mechanism requires that  $\omega$  be fixed externally by the worldtube closure condition  $\omega T = 2\pi n$ , not by spatial relaxation. This is the case for the rotating-frame-periodic BVP solver (PeriodicBC/solve\_block): the JFNK root-find relaxes node positions only;  $\omega$  is not a degree of freedom. Running the vortex through the breather eigensolver (where  $\omega$  co-relaxes as a nonlinear eigenvalue) would violate this condition and revert the binding verdict to net-repulsive.

**Binding-gravity co-variation.** The same  $r_t$  (with  $\alpha_t = \alpha$ ) sourcing the binding compression (15) also sources gravitational time dilation (Paper II): both coupling strengths scale as  $\propto \alpha$ . As a result, binding strength and gravitational time-dilation strength co-vary in  $\alpha$  and are not independently tunable. If a simulation can tune binding without tuning gravity, the identification  $\alpha_t = \alpha$  is falsified.

#### 4.5. Open normalization problem: Compton wavelength

The mass-energy identification (7) requires three open normalization steps that are not completed here:

1. **Explicit toroidal energy integral.** Compute  $E_2$ ,  $E_4$ , and  $E_{\text{curv}}$  in (12) explicitly by integrating the substrate energy density over the torus geometry with the hedgehog-like core profile of Section 3.
2. **Equilibrium amplitude.** The amplitude  $A$  is not fixed by the Derrick balance alone; it is selected by the energy normalization  $E(R^*, A) = mc^2$  for a chosen target particle species. Express  $A$  in terms of substrate parameters  $(k_s, \alpha, a, m_{\text{node}})$  and the target species mass.
3. **Compton wavelength connection.** Show that  $R^* = \hbar/mc$  using consistent derivations of  $\hbar$  and  $c$  from substrate parameters. Because  $c^2 = k_s a^2 / m_{\text{node}}$  and the natural action quantum is set by  $k_s a^3$  (energy times volume) or  $k_s a^2 \cdot (a/c)$  (energy times time), the combination giving  $\hbar$  in SI units must close the triangle  $R^* \cdot mc = \hbar$  without additional free parameters.

**Inertial mass and the classical 4/3 caveat.** Equation (7) is an energy definition of mass; a consistent theory must reproduce the inertial mass — the substrate momentum carried per unit drift velocity,  $p = mv$  — with the same  $m$ . That a traveling excitation of an elastic masses-and-springs medium carries real momentum proportional to its energy is established generally by Peskin [11], who finds  $\bar{P} = \bar{\mathcal{E}}/c$  for acoustic waves as a second-order effect of the density–velocity correlation; the same mechanism gives the inertia of the confined toroidal mode here. The classical electromagnetic-mass programme is a cautionary precedent: weighing only the field energy of a charged sphere gives an inertial mass  $\frac{4}{3}U/c^2$  that disagrees with the energy mass  $U/c^2$  (the “4/3 problem”), because a pure field lump is unstable and constitutes an open subsystem. The discrepancy vanishes only when the non-electromagnetic binding stresses that hold the lump together are included in the stress–energy tensor, restoring a proper four-vector and  $m_{\text{inertial}} = E_{\text{rest}}/c^2$  [8,9,12,13]. In the present model the geometric quartic anti-expansion ( $\propto k_s \alpha / a$ ) and the time-link binding compression (15) play the role of Poincaré stresses. Open problem 1 below is therefore also the 4/3 test: the explicit toroidal energy integral must include these stresses so that  $E(R^*) = mc^2$  and the moving-soliton momentum yield a single  $m$ .

#### 4.6. Mass bridge open problems

1. **Explicit toroidal energy and equilibrium radius.** Complete the calculation in Section 4.3 for the substrate spring law (Equation (20) of Paper I). Identify  $T_{\text{string}}$  and  $E_{\text{curv}}$  in terms of  $(k_s, \alpha, a, w, \omega_0)$ . The integral must include the geometric quartic and time-link binding stresses so that the resulting  $E(R^*) = mc^2$  and the moving-soliton momentum  $p = mv$  yield the same  $m$  (the 4/3 consistency test).
2. **Verify binding via closure-locked  $\omega$  sweep.** Solve a stationary worldtube at each closure index  $n$  (each imposed  $\omega = 2\pi n / (P\Delta t)$ ); on each converged state measure  $\Delta u_{\parallel}(\rho)$  and the relaxed amplitude  $A$ . Confirm  $\Delta u_{\parallel} \propto \omega^2$  across the converged family (dividing out  $A$ , which self-selects to the imposed  $\omega$ ).
3. **Winding-lock test.** Vary the topological winding  $B$  at fixed amplitude on a converged baryon texture; check whether the P-odd holonomy  $\gamma_T \propto B$  (soft winding-lock) or is flat in  $B$  (energetic-only). A  $B$ -slope of 1 confirms charge tracks winding; flat-in- $B$  indicates the energetic leg only.
4. **Compton wavelength derivation.** Carry out the three-step normalization of Section 4.5 and test whether  $R^* = \hbar/mc$  closes without additional free parameters.



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**Data Availability Statement:** The substrate solver, diagnostic suite, and all experiments reported in this series are openly available at <https://github.com/LukasMolzberger/BraneSim> [14].

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## Appendix A. Symbol dictionary

|                                        |                                                                                        |
|----------------------------------------|----------------------------------------------------------------------------------------|
| $X(x, t) \in \mathbb{R}^4$             | Embedding field / 4D brane configuration (Paper I).                                    |
| $t$                                    | Slicing parameter along the timelike direction.                                        |
| $\alpha$                               | Dimensionless pre-stress parameter: $\alpha = \ell_0/h_* \in (0, 1]$ (Paper I).        |
| $\epsilon$                             | Normalized polarization state of a narrowband wave; angular structure of a VSH mode.   |
| $\mathcal{A}_\mu$                      | Wilczek–Zee / gauge connection on the polarization bundle (Paper III).                 |
| $\gamma_\Gamma$                        | Geometric phase / holonomy accumulated around a closed loop $\Gamma$ in field space.   |
| $\Psi = (\psi_x, \psi_y, \psi_z)^\top$ | Axis-aligned complex narrowband triplet carrier; soliton amplitude envelope.           |
| $k_s$                                  | Central-force spring stiffness of the 6-neighbor stencil (Paper I).                    |
| $m$                                    | Node mass (discrete lattice; continuum mass density $\rho_m = m/a^3$ ).                |
| $\rho_m$                               | Brane mass density.                                                                    |
| $a$                                    | Lattice spacing of the cubic substrate (microphysical length scale).                   |
| $\Delta t$                             | Temporal lattice step; period of the worldtube closure condition $\omega T = 2\pi n$ . |
| $c_L$                                  | Long-wavelength longitudinal wave speed: $c_L^2 = k_s a^2 / m$ (Paper I).              |

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